

# Atoms

We now proceed towards proving a *representation theorem* for the class of Boolean algebras. For this, let us go through the basic notion and properties of *atoms* in these structures.

**Definition 0.1. (Atom)** An element  $a$  in a Boolean algebra  $(B, \leq, \vee, \wedge, ^c, 0, 1)$  is called an *atom* if and only if  $a \neq 0$ , and for all  $b \in B$ , if  $b \leq a$ , then either  $b = a$  or  $b = 0$ .

The Boolean algebra is said to be *atomic* if and only if for every  $a(\neq 0) \in B$ , there exists an atom  $b(\in B)$  such that  $b \leq a$ .

The algebra is *atomless*, if there are no atoms in it.

**Example 0.1.**

1. In the power set Boolean algebra of a set  $U$ , viz.  
 $(\mathcal{P}(U), \cup, \cap, ^c, \emptyset, U)$ ,  
the atoms are exactly the singletons.  
The algebra is clearly atomic.
2. Consider the Boolean algebra  
 $(\mathcal{F}/\equiv, \leq, \vee, \wedge, \neg, 0, 1)$ , where  $PV := \{p_1, \dots, p_n\}$ .
  - (a) For each  $n$ -tuple  
 $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)$  in  $\{0, 1\}^n$ ,  $[\bigwedge_{1 \leq i \leq n} \varepsilon_i p_i]$  is an atom.
  - (b) Moreover, every atom in the algebra must be of this form.  
So, for instance, if  $PV := \{p_1, p_2\}$ , the atoms of the corresponding Boolean algebra  $(\mathcal{F}/\equiv, \leq, \vee, \wedge, \neg, 0, 1)$  are exactly the four elements  $[p_1 \wedge p_2]$ ,  $[p_1 \wedge \neg p_2]$ ,  $[\neg p_1 \wedge p_2]$ ,  $[\neg p_1 \wedge \neg p_2]$ .
  - (c) The algebra is atomic.
3. The Boolean algebra  
 $(\mathcal{F}/\equiv, \leq, \vee, \wedge, \neg, 0, 1)$ , where  $PV$  is *infinite*,  
is atomless.

*Proof.* 1 is obvious.

For 2, note that we need to prove the following.

(a) For any  $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n) \in \{0, 1\}^n$ ,

(i)  $\bigwedge_{1 \leq i \leq n} \varepsilon_i p_i \neq \perp$ ;

(ii) if  $\alpha \neq \perp$  and  $\models \alpha \rightarrow \bigwedge_{1 \leq i \leq n} \varepsilon_i p_i$  then  $\alpha \equiv \bigwedge_{1 \leq i \leq n} \varepsilon_i p_i$ .

These are left as exercises.

(b) For any  $\alpha \neq \perp$ , we know that

$\alpha \equiv \bigvee_{(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n) \in X} (\bigwedge_{1 \leq i \leq n} \varepsilon_i p_i)$ ,

the CDNF of  $\alpha$  ( $X$  is a non-empty subset of  $\{0, 1\}^n$ ). Then

$\models (\bigwedge_{1 \leq i \leq n} \varepsilon_i p_i) \rightarrow \alpha$

for each  $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n) \in X$ .

So if  $[\alpha]$  is an atom, it is clear that

$\alpha \equiv \bigwedge_{1 \leq i \leq n} \varepsilon_i p_i$ , for each

$(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n) \in X$ .

This means that  $X$  must be a singleton.

(c) Similar to (b):

any  $\alpha \neq \perp$  is logically equivalent to its CDNF, of the form

$\bigvee_{(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n) \in X} (\bigwedge_{1 \leq i \leq n} \varepsilon_i p_i)$ ,

$X$  a non-empty subset of  $\{0, 1\}^n$ .

Then  $[\bigwedge_{1 \leq i \leq n} \varepsilon_i p_i]$ , for each  $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n) \in X$ ,

serves as an atom below  $[\alpha]$ .

3 requires the following to be proved.

For each wff  $\alpha \neq \perp$ , there is a wff  $\beta$  such that

(a)  $\beta \neq \perp$ ,

(b)  $\models \beta \rightarrow \alpha$ , but

(c)  $\beta \neq \alpha$ .

As  $\alpha$  is a finite string, it can only involve a finite number of propositional variables from  $PV$ .

We consider  $\beta := \alpha \wedge p$ , where  $p \in PV$  does not occur in  $\alpha$ .

We have a choice for  $p$ , as  $PV$  is infinite in this case.

It is straightforward to verify (a)-(c) for such a  $\beta$ . □

**Proposition 0.1.** Every finite Boolean algebra is atomic.

*Proof.* Let  $a (\neq 0)$  in a finite Boolean algebra

$(B, \leq, \vee, \wedge, ^c, 0, 1)$ . If  $a$  is an atom, we are done.

Suppose  $a$  is not an atom. Then there is at least one non-zero element of  $B$  strictly below  $a$ . This means that the set

$L(a) := \{x \in B : 0 \neq x \leq a \text{ and } x \neq a\}$  is non-empty.

But then finiteness of  $B$  (and hence of  $L(a)$ ) guarantees the existence of a *minimal* element, say  $m$ , in  $L(a)$ .  $m$  would serve as an atom below  $a$ . □

Observe that we could use Proposition 0.1 to conclude that the algebra in Example 0.1(2) is atomic:  
we know that for finite  $PV$ , the quotient set  $\mathcal{F}/\equiv$  is finite.

The following gives a special property of any atom in a Boolean algebra  $(B, \leq, \vee, \wedge, ^c, 0, 1)$ , and we shall use it in the next lecture, in the proof of the representation theorem.

**Exercise 0.1.** If  $a$  is an atom, then for all  $x \in B$ , either  $a \leq x$  or  $a \leq x^c$ .